Experiment 8

Building anonymous FTP Scanner using ftplib module.

Aim:

To build an anonymous FTP Scanner using python's ftplib module to identify open FTP servers that allows anonymous login.

Theory

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server over TCP/IP.

Procedure:

1. import ftplib, socket and concurrent.futures modules.
2. Define a function anon_ftp_scan that attempts anonymous FTP login.
3. Use ftplib.FTP() with a timeout to connect to a target IP, port 21.
4. If login succeeds list the root directory using ftp.nlst()

Command / Code:

```
import ftplib, socket
from concurrent.futures import ThreadPoolExecutor

def anon_ftp_scan(ip):

  try:
    ftp = ftplib.FTP(timeout=3)
    ftp.connection(ip, 21, timeout=3)
   ftp.login['anonymous', 'anon@test.com')
  print(f'[+] {ip} - Anonymous OPEN')
   files = ftp.nlst()
  print(f'[-] {files[:5]}')
  ftp.quit()
  return true
```

5. Print the IP, welcome message, and the available files.

6. Handle exception ConnectionRefused Error, error socket. timeout.

7. Create a list of IP address to scan [use lab network range)

8. Run the scanner and recieved results

9. use ftplib quit() to Properly close each connection.

$\frac{12}{15} \times \frac{9}{10}$

Result:

The anonymous FTP server was implemented. Successfully open FTP server allowing anonymous login were identified and directory listings were retrieved from accessible sources.